
TRAVAUX DIRIG  S N   4 : Risk Convexification

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EXERCICE 1. We work in the framework of binary classification : we consider a random descriptor X with distribution P_X taking values in $\mathbb{R}^d$ ($d \in \mathbb{N}^*$), equipped with its Borel sigma-algebra, and a random label Y taking the value -1 or 1 .

The regression function (posterior probability) is denoted by $\eta : x \in \mathbb{R}^d \mapsto \mathbb{P}(Y = 1 \mid X = x) \in [0, 1]$. We assume that $0 < \eta(X) < 1$ almost surely.

Let $\mathcal{G} := \{g : \mathbb{R}^d \rightarrow \{-1, 1\}\}$ be the set of classifiers suited to this setting. The classification error is defined as the map $L : g \in \mathcal{G} \mapsto \mathbb{P}(Y \neq g(X)) \in [0, 1]$, and we write $L^* := \inf_{g \in \mathcal{G}} L(g)$.

Let $\mathcal{F} := \{f : \mathbb{R}^d \rightarrow \mathbb{R}\}$ and $\text{sgn} : a \in \mathbb{R} \mapsto 2 \mathbb{1}_{\{a \geq 0\}} - 1 \in \{-1, 1\}$. In this exercise we focus specifically on the set of classifiers $\mathcal{G}_{\mathcal{F}} := \{\text{sgn} \circ f : f \in \mathcal{F}\}$.

Finally, let $\phi : \mathbb{R} \rightarrow \mathbb{R}_+$ be a differentiable, strictly convex, nondecreasing function satisfying $\lim_{x \rightarrow -\infty} \phi(x) = 0$ and $\phi(0) = 1$. For every $f \in \mathcal{F}$ (and hence every classifier in $\mathcal{G}_{\mathcal{F}}$) we consider the convexified version of the classification risk :

$$A(f) := \mathbb{E} \left[\phi(-Yf(X)) \right].$$

- 1) For every $u \in [0, 1]$ define $h_u : a \in \mathbb{R} \mapsto u \phi(-a) + (1 - u) \phi(a)$. Show that $\min_{f \in \mathcal{F}} A(f)$ is attained at

$$f^* : x \in \mathbb{R}^d \mapsto \arg \min_{a \in \mathbb{R}} h_{\eta(x)}(a). \quad (1)$$

Is this function well defined for every $x \in \mathbb{R}^d$?

- 2) Differentiate the function $h_{\eta(x)}$ for an arbitrary $x \in \mathbb{R}^d$, then check that $\text{sgn} \circ f^*$ coincides with the Bayes classifier.
- 3) (Zhang's lemma) Let $H : u \in]0, 1[\mapsto \min_{a \in \mathbb{R}} h_u(a)$ and assume there exist constants $s > 1$ and $c > 0$ such that

$$\forall u \in]0, 1[, \quad \left| \frac{1}{2} - u \right|^s \leq c^s (1 - H(u)). \quad (2)$$

- a) Let $f \in \mathcal{F}$. Show that $L(\text{sgn} \circ f) - L^* \leq 2 \left(\mathbb{E} \left[|\eta(X) - 1/2|^s \mathbb{1}_{\{(2\eta(X)-1)f(X) < 0\}} \right] \right)^{1/s}$.

- b) Let $A^* := \inf_{f \in \mathcal{F}} A(f)$. Deduce from the previous question that for every function $f \in \mathcal{F}$,

$$L(\text{sgn} \circ f) - L^* \leq 2c (A(f) - A^*)^{1/s}. \quad (3)$$

- c) What is H when $\phi = \exp$? For which constants s and c does condition (2) hold?

Correction Exercise 1

1) Let $f \in \mathcal{F}$. We have :

$$\begin{aligned} A(f) &= \mathbb{E} [\phi(-Yf(X))] = \mathbb{E} [\phi(-f(X))\mathbb{1}_{\{Y=1\}} + \phi(f(X))\mathbb{1}_{\{Y=-1\}}] \\ &= \mathbb{E} [\phi(-f(X))\eta(X) + \phi(f(X))(1 - \eta(X))] \quad (\text{total expectation, conditioning on } X) \\ &= \mathbb{E} [h_{\eta(X)}(f(X))] = \int_{\mathbb{R}^d} h_{\eta(x)}(f(x)) \mathbf{P}_X(dx). \end{aligned}$$

Since $h_{\eta(x)}(f(x)) \geq \min_{a \in \mathbb{R}} h_{\eta(x)}(a) = h_{\eta(x)}(f^*(x))$ for every x , by construction, if f^* is well defined for $\mathbf{P}_X$ -almost every x , then $A(f) \geq A(f^*)$ for all $f \in \mathcal{F}$, i.e. f^* minimizes the convexified risk.

Let us now check this condition. Recall the assumptions on $\phi : \mathbb{R} \rightarrow \mathbb{R}_+$:

- (i) ϕ is differentiable,
- (ii) ϕ is strictly convex,
- (iii) ϕ is nondecreasing,
- (iv) $\lim_{x \rightarrow -\infty} \phi(x) = 0$,
- (v) $\phi(0) = 1$.

Note that $x \mapsto e^x$ satisfies all these assumptions. We deduce several additional properties :

- (vi) ϕ is strictly increasing.

Proof : ϕ is nondecreasing ; if it were not strictly increasing, it would be constant on some interval, which contradicts strict convexity.

- (vii) $\phi > 0$.

Proof : immediate consequence of (iv) and (vi) : for every x , $\phi(x) > \phi(x-1) \geq \lim_{-\infty} \phi = 0$.

- (viii) $\phi'(0) > 0$.

Proof : ϕ being strictly convex and differentiable, for every $x \neq 0$ we have $\phi(x) > \phi(0) + \phi'(0)x$ (the graph of ϕ lies strictly above its tangent at 0). Since ϕ is nondecreasing, $\phi'(0) \geq 0$. By contradiction, if $\phi'(0) = 0$, then for every $x < 0$ we would get $\phi(x) > \phi(0)$, which contradicts the monotonicity of ϕ . Hence $\phi'(0) > 0$.

- (ix) $\lim_{x \rightarrow +\infty} \phi(x) = +\infty$.

Proof : as before, for every $x \neq 0$, $\phi(x) > \phi(0) + \phi'(0)x$, and using (viii) we get $\lim_{x \rightarrow +\infty} \phi(x) = +\infty$.

Now let $u \in [0, 1]$ and let us study the function h_u .

- **Case** $u \in \{0, 1\}$. Properties (iv) and (vi) imply that ϕ has no minimum, only an infimum (equal to 0) reached at $-\infty$. Hence $h_0 : a \mapsto \phi(a)$ and $h_1 : a \mapsto \phi(-a)$ have no minimum, and f^* is not defined on the set $\mathcal{X}_0 := \{x \in \mathbb{R}^d : \eta(x) \in \{0, 1\}\}$.

- **Case** $u \in]0, 1[$. We first note that :

- (x) $\lim_{a \rightarrow +\infty} h_u(a) = \lim_{a \rightarrow -\infty} h_u(a) = +\infty$, by (iv) and (ix) (and since $u > 0$ and $1 - u > 0$),
- (xi) $h_u(0) = 1$, by (v),
- (xii) h_u is continuous, since ϕ , and thus h_u , is differentiable.

By (x) and (xi), there exists $\alpha > 0$ such that $h_u(a) > 1 = h_u(0)$ for every $a \in]-\infty, -\alpha[\cup]\alpha, +\infty[$. By (xii), h_u is continuous on the compact $[-\alpha, \alpha]$, so it is bounded and attains its bounds (extreme value theorem) : it has at least one minimizer m_u on $[-\alpha, \alpha]$. Moreover, ϕ being strictly convex, so is h_u (positive combination of strictly convex functions),

hence m_u is unique. By construction, $h_u(m_u) \leq 1 = h_u(0) < h_u(a)$ for $|a| > \alpha$, so m_u is in fact the unique global minimizer of h_u on $\mathbb{R}$.

Therefore, f^* is well defined on the set $\mathcal{X}_1 := \{x \in \mathbb{R}^d : \eta(x) \in]0, 1[\}$.

Finally, by assumption, $\eta(X) \in]0, 1[$ $\mathbf{P}_X$ -a.s., i.e. $\mathbf{P}_X(\mathcal{X}_0) = 0$ and $\mathbf{P}_X(\mathcal{X}_1) = 1$. We conclude that f^* is well defined for $\mathbf{P}_X$ -almost every $x \in \mathbb{R}^d$ (which is enough since A only depends on the values of f $\mathbf{P}_X$ -a.s.).

2) Let $x \in \mathbb{R}^d$. Since ϕ is differentiable (i), so is $h_{\eta(x)} : \mathbb{R} \rightarrow \mathbb{R}_+$, and for every $a \in \mathbb{R}$:

$$h'_{\eta(x)}(a) = -\eta(x)\phi'(-a) + (1 - \eta(x))\phi'(a).$$

Let us check that $\text{sgn} \circ f^*$ coincides with the Bayes classifier $g^* : x \mapsto 2\mathbb{1}_{\{\eta(x) \geq 1/2\}} - 1$. Let $x \in \mathcal{X}_1$. We have seen that $h_{\eta(x)}$ is differentiable, strictly convex, and has a unique minimizer at $f^*(x)$. Therefore, $h'_{\eta(x)}(a) = 0 \iff a = f^*(x)$. Now, $h'_{\eta(x)}(a) = 0$ means :

$$\frac{\eta(x)}{1 - \eta(x)} = \frac{\phi'(a)}{\phi'(-a)}.$$

(Since $x \in \mathcal{X}_1$, $1 - \eta(x) \neq 0$. Moreover, ϕ being strictly convex and nondecreasing, its derivative is nonnegative and strictly increasing on $\mathbb{R}$, hence strictly positive.) We thus have $\frac{\eta(x)}{1 - \eta(x)} = \frac{\phi'(f^*(x))}{\phi'(-f^*(x))}$, and :

$$\begin{aligned} \eta(x) > \frac{1}{2} &\iff \frac{\eta(x)}{1 - \eta(x)} > 1 \iff \frac{\phi'(f^*(x))}{\phi'(-f^*(x))} > 1 \iff \phi'(f^*(x)) > \phi'(-f^*(x)) \\ &\iff f^*(x) > -f^*(x) \iff f^*(x) > 0 \quad (\phi' \text{ strictly increasing}). \end{aligned}$$

The same reasoning with equalities gives $\eta(x) = \frac{1}{2} \iff f^*(x) = 0$, hence $\eta(x) \geq \frac{1}{2} \iff f^*(x) \geq 0$. Finally :

$$\text{sgn} \circ f^*(x) = 2\mathbb{1}_{\{f^*(x) \geq 0\}} - 1 = 2\mathbb{1}_{\{\eta(x) \geq \frac{1}{2}\}} - 1 = g^*(x),$$

for $\mathbf{P}_X$ -almost every $x \in \mathbb{R}^d$. The function $\text{sgn} \circ f^*$ indeed coincides with the Bayes classifier.

3) a) Let $f \in \mathcal{F}$ and $g = \text{sgn} \circ f$. We have seen in exercise 2 of TD2 that :

$$L(g) - L^* = 2\mathbb{E} \left[\left| \eta(X) - \frac{1}{2} \right| \mathbb{1}_{\{g(X) \neq g^*(X)\}} \right] = 2\mathbb{E} \left[\left| \eta(X) - \frac{1}{2} \right| \left| \mathbb{1}_{\{f(X) < 0\}} - \mathbb{1}_{\{\eta(X) < \frac{1}{2}\}} \right| \right].$$

Now, for every $x \in \mathbb{R}^d$ such that $\eta(x) \neq \frac{1}{2}$ (the other points do not contribute to the expectation) :

$$\left| \mathbb{1}_{\{f(x) < 0\}} - \mathbb{1}_{\{\eta(x) < \frac{1}{2}\}} \right| = \begin{cases} 0 & \text{if } \text{sgn}(f(x)) = \text{sgn}(2\eta(x) - 1) \\ 1 & \text{otherwise} \end{cases} \leq \mathbb{1}_{\{(2\eta(x) - 1)f(x) \leq 0\}}.$$

Hence :

$$\begin{aligned} L(\text{sgn} \circ f) - L^* &\leq 2\mathbb{E} \left[\left| \eta(X) - \frac{1}{2} \right| \mathbb{1}_{\{(2\eta(X) - 1)f(X) \leq 0\}} \right] \\ &= 2\mathbb{E} \left[\left(\left| \eta(X) - \frac{1}{2} \right| \mathbb{1}_{\{(2\eta(X) - 1)f(X) \leq 0\}} \right)^s \right]^{\frac{1}{s}} \\ &\leq 2 \left(\mathbb{E} \left[\left| \eta(X) - \frac{1}{2} \right|^s \mathbb{1}_{\{(2\eta(X) - 1)f(X) \leq 0\}} \right] \right)^{\frac{1}{s}}, \end{aligned}$$

using Jensen's inequality for the concave function $x \in \mathbb{R}_+ \mapsto x^{1/s}$ (concave since $s > 1$).

Remark : the indicator must be taken with a large inequality $(2\eta(X) - 1)f(X) \leq 0$. Indeed, with the convention $\text{sgn}(0) = 1$, if $f(x) = 0$ and $\eta(x) < \frac{1}{2}$, then $\text{sgn}(f(x)) = 1 \neq -1 = g^*(x)$ is a mistake, whereas $(2\eta(x) - 1)f(x) = 0$. For instance, with $f \equiv 0$ and $\eta < \frac{1}{2}$ everywhere, we get $L(\text{sgn} \circ f) - L^* = \mathbb{E}[1 - 2\eta(X)] > 0$, while the bound with a strict inequality would be equal to 0.

b) By assumption, there exist $s > 1$ and $c > 0$ such that for every $u \in]0, 1[$:

$$\left| \frac{1}{2} - u \right|^s \leq c^s (1 - H(u)).$$

Therefore, for every $x \in \mathcal{X}_1$:

$$\left| \eta(x) - \frac{1}{2} \right|^s \leq c^s (1 - H(\eta(x))),$$

with $H(\eta(x)) = \min_{a \in \mathbb{R}} h_{\eta(x)}(a) = h_{\eta(x)}(f^*(x))$ by question 1.

Let $f \in \mathcal{F}$. Plugging this inequality into the result of question 3.a gives :

$$\begin{aligned} L(\text{sgn} \circ f) - L^* &\leq 2 \left(\mathbb{E} \left[c^s (1 - h_{\eta(X)}(f^*(X))) \mathbb{1}_{\{(2\eta(X)-1)f(X) \leq 0\}} \right] \right)^{\frac{1}{s}} \\ &= 2c \left(\mathbb{E} \left[(1 - h_{\eta(X)}(f^*(X))) \mathbb{1}_{\{(2\eta(X)-1)f(X) \leq 0\}} \right] \right)^{\frac{1}{s}}. \end{aligned}$$

Since $x \in \mathbb{R}_+ \mapsto x^{1/s}$ is nondecreasing and $c > 0$, it remains to show that $A(f) - A^*$ upper bounds this last expectation.

We have seen in question 1 that $A^* = A(f^*)$, so $A(f) - A^* = \mathbb{E} [h_{\eta(X)}(f(X)) - h_{\eta(X)}(f^*(X))]$. Let $x \in \mathcal{X}_1$.

– If $(2\eta(x) - 1)f(x) > 0$, then by definition of f^* :

$$(1 - h_{\eta(x)}(f^*(x))) \mathbb{1}_{\{(2\eta(x)-1)f(x) \leq 0\}} = 0 \leq h_{\eta(x)}(f(x)) - h_{\eta(x)}(f^*(x)).$$

– If $(2\eta(x) - 1)f(x) \leq 0$, then :

$$\begin{aligned} h_{\eta(x)}(f(x)) &= \eta(x)\phi(-f(x)) + (1 - \eta(x))\phi(f(x)) && \text{(definition of } h_{\eta(x)}) \\ &\geq \phi(-\eta(x)f(x) + (1 - \eta(x))f(x)) && (\phi \text{ convex}) \\ &= \phi(-f(x)(2\eta(x) - 1)) \\ &\geq \phi(0) && (\phi \text{ nondecreasing and } -f(x)(2\eta(x) - 1) \geq 0) \\ &= 1 && \text{(by (v)),} \end{aligned}$$

hence :

$$h_{\eta(x)}(f(x)) - h_{\eta(x)}(f^*(x)) \geq 1 - h_{\eta(x)}(f^*(x)) = (1 - h_{\eta(x)}(f^*(x))) \mathbb{1}_{\{(2\eta(x)-1)f(x) \leq 0\}}.$$

We have thus shown that, $\mathbb{P}_X$ -a.s. :

$$h_{\eta(X)}(f(X)) - h_{\eta(X)}(f^*(X)) \geq (1 - h_{\eta(X)}(f^*(X))) \mathbb{1}_{\{(2\eta(X)-1)f(X) \leq 0\}},$$

and the inequality remains true when taking the expectation on both sides. Finally :

$$L(\text{sgn} \circ f) - L^* \leq 2c (A(f) - A^*)^{1/s}.$$

c) Let $\phi : x \in \mathbb{R} \mapsto e^x$ and $u \in]0, 1[$. Then, for every $a \in \mathbb{R}$, $h_u(a) = ue^{-a} + (1-u)e^a$, and $h'_u(a) = -ue^{-a} + (1-u)e^a$. Thus h_u attains its minimum at $a_u \in \mathbb{R}$ such that :

$$\frac{u}{1-u} = \frac{e^{a_u}}{e^{-a_u}} = e^{2a_u}, \quad \text{i.e.} \quad a_u = \frac{1}{2} \ln \left(\frac{u}{1-u} \right),$$

and :

$$h_u(a_u) = u\sqrt{\frac{1-u}{u}} + (1-u)\sqrt{\frac{u}{1-u}} = 2\sqrt{u}\sqrt{1-u}.$$

The function H is thus $H : u \in]0, 1[\mapsto 2\sqrt{u(1-u)}$. It is differentiable, with derivative $H' : u \in]0, 1[\mapsto \frac{1-2u}{\sqrt{u(1-u)}}$, leading to the following table of variations :

u	0	$\frac{1}{2}$	1		
$H'(u)$		+	0	-	
$H(u)$	0	$\nearrow$	1	$\searrow$	0

Hence, for every $u \in]0, 1[$, $\sqrt{u}\sqrt{1-u} \leq \frac{1}{2}$, and :

$$\begin{aligned} 1 - H(u) &= 1 - 2\sqrt{u}\sqrt{1-u} = 2 \left(\frac{1}{2} - \sqrt{u}\sqrt{1-u} \right) \\ &\geq 2 \left(\frac{1}{2} - \sqrt{u}\sqrt{1-u} \right) \left(\frac{1}{2} + \sqrt{u}\sqrt{1-u} \right) \quad (\text{the last factor is } \leq 1, \text{ the first one is } \geq 0) \\ &= 2 \left(\frac{1}{4} - u(1-u) \right) = 2 \left(\frac{1}{4} + u^2 - u \right) \\ &= 2 \left(\frac{1}{2} - u \right)^2, \end{aligned}$$

which finally gives :

$$\left| \frac{1}{2} - u \right|^2 \leq \frac{1}{2} (1 - H(u)).$$

The constants $s = 2$ and $c = \frac{1}{\sqrt{2}}$ thus satisfy condition (2). For the exponential loss (used in AdaBoost), we then get :

$$L(\text{sgn} \circ f) - L^* \leq \sqrt{2} (A(f) - A^*)^{1/2}.$$